

Valleys & Their Types

A GEOGRAPHY OF LANDFORMS

From the mighty gorges carved by rivers to the sweeping troughs sculpted by glaciers — valleys are among Earth's most dramatic and diverse landforms.



GEOGRAPHY



LANDFORMS

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What Is a Valley?

DEFINITION

A valley is a low-lying area of land situated between hills, mountains, or other elevated landforms. Valleys are typically elongated and are commonly carved by the erosive action of rivers, glaciers, or tectonic forces acting over thousands to millions of years.

FORMATION AGENTS

Rivers, glaciers, wind, and tectonic activity are the primary forces that shape valleys over geological timescales.

IMPORTANCE

Valleys provide fertile land, water sources, and shelter — making them cradles of human civilisation throughout history.



GEOGRAPHY

U-Shaped Valley (Glacial Trough)

GLACIER-CARVED
VALLEY

FORMATION

Carved by the immense erosive power of glaciers that scour both the floor and sides of a valley equally, producing a broad, flat bottom and steep sides shaped like the letter "U".

CHARACTERISTICS

Wide, flat floor; near-vertical walls; often contain glacial lakes (ribbon lakes); highly fertile after deglaciation.

EXAMPLE

Yosemite Valley in California, and the fjords of Norway and New Zealand.

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Gorge & Canyon

GORGE

A gorge is a deep, narrow valley with very steep or vertical rocky walls, formed when a river rapidly cuts through hard rock. Gorges are narrower and often shorter than canyons. Example: Borra Caves Gorge, Andhra Pradesh.

CANYON

A canyon is a large gorge — deeper, wider, and longer — typically formed in arid or semi-arid regions where rivers cut through soft sedimentary rock over millions of years. Example: Grand Canyon, USA (1.6 km deep).

FORMATION PROCESS

Both are formed primarily by river downcutting combined with uplift of the surrounding terrain. Lack of vegetation in dry climates allows deeper cutting without lateral erosion widening the valley.

KEY DIFFERENCE

A gorge is a steep-sided, narrow cut. A canyon is larger in scale — often featuring stepped walls, side ravines, and layered rock strata visible in cross-section.

TECTONICS

Rift Valley



FORMATION

Formed when tectonic plates pull apart (diverge), causing the land between two parallel faults to sink downward, creating a long, linear depression called a graben.

CHARACTERISTICS

Very long (hundreds to thousands of km); flat or depressed floor; volcanic activity nearby; often contains lakes and rivers.

EXAMPLE

The Great Rift Valley of East Africa stretches over 6,000 km; the Rhine Rift Valley in Europe; and the Jordan Rift Valley in the Middle East.

6,000+
KM LONG (E. AFRICA)

30+
LAKES FORMED

~30M
YEARS OLD

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Hanging Valley

WHAT IS A HANGING VALLEY?

A hanging valley is a tributary valley that is perched high above the floor of the main valley. This happens because the main glacier erodes far more deeply than the smaller tributary glacier, leaving the tributary valley "hanging" above the main trough.

KEY FEATURE

Where a hanging valley meets the main valley, there is typically a dramatic waterfall or steep cascade — one of the most spectacular landforms in glaciated regions.

EXAMPLE

Bridalveil Fall in Yosemite (USA) and the many waterfalls of Milford Sound, New Zealand, are formed from hanging valleys above glacial troughs.

 GEOGRAPHY

Blind Valley & Box Canyon

BLIND VALLEY

A valley that ends abruptly at a steep rock wall with no surface outlet. A river enters the valley but disappears underground through a cave or sinkhole, common in karst limestone landscapes.

EXAMPLE – BLIND VALLEY

Rakov Škocjan, Slovenia; many karst valleys across the Dinaric Alps where rivers disappear into underground cave systems.

BOX CANYON

A canyon with three steep or vertical rock walls and one open end through which a river enters. The closed end (head) is sheer rock. Often formed by headward erosion.

EXAMPLE – BOX CANYON

Box Canyon Falls, Colorado, USA; several box canyons in the Colorado Plateau region formed in sandstone and granite rock.

Flat-Floored & Wide Valley



FORMATION

As rivers mature and their gradient decreases, lateral (sideways) erosion becomes dominant over downcutting. This gradually widens the valley floor, creating a broad, flat plain called a floodplain.

CHARACTERISTICS

Wide, flat floor; meandering river; fertile alluvial soil deposited during floods; often densely populated due to agricultural potential.

EXAMPLE

The Indo-Gangetic Plain, the Nile Valley in Egypt, and the Mississippi River valley in the United States.



Dry & Wind-Carved Valleys

DRY VALLEY

A valley with no permanent river. Originally carved by water (glacial meltwater or past rivers), these valleys were left dry as climate changed. Common in chalk downlands of England and semi-arid regions.

EXAMPLE – DRY VALLEY

The Coombes of the South Downs, England; many dry valleys in the Negev Desert, Israel, formed by ancient flash floods.

WIND-CARVED (DEFLATION) VALLEY

In extremely arid deserts, strong winds strip away loose sediment and carve out shallow, elongated depressions over thousands of years — a process called deflation or aeolian erosion.

EXAMPLE – WIND VALLEY

The Qattara Depression, Egypt (134 m below sea level) is one of the world's largest wind-scoured depressions, shaped largely by aeolian forces.



SIGNIFICANCE

Why Valleys Matter



HUMAN & ECONOMIC VALUE

- ◆ Fertile alluvial soil supports agriculture
- ◆ Rivers provide freshwater for irrigation
- ◆ Natural corridors for trade & transport
- ◆ Sites of ancient civilisations (Indus, Nile)
- ◆ Hydropower generation from river falls



ECOLOGICAL VALUE

- ◆ Biodiversity hotspots with varied habitats
- ◆ Groundwater recharge zones
- ◆ Forest and wildlife corridors
- ◆ Climate regulation through vegetation
- ◆ Tourism: trekking, rafting, scenic landscapes



Valleys are not just landforms — they are the geographic foundations of human civilisation, ecological systems, and economic activity across every continent.

Valley Types — At a Glance

VALLEY TYPE	FORMED BY	SHAPE	NOTABLE EXAMPLE
V-SHAPED	River erosion (vertical)	Narrow, steep sides	Upper Ganges, Himalayas
U-SHAPED	Glacial erosion	Wide, flat floor	Yosemite Valley, USA
GORGE / CANYON	River downcutting	Deep, narrow walls	Grand Canyon, USA
RIFT VALLEY	Tectonic plate divergence	Long, linear depression	Great Rift Valley, Africa
HANGING VALLEY	Unequal glacial erosion	Tributary above main floor	Bridalveil Fall, Yosemite
FLAT-FLOORED	Lateral river erosion	Broad with floodplain	Indo-Gangetic Plain
BLIND / BOX CANYON	River / karst erosion	Closed at one/both ends	Rakov Škocjan, Slovenia
DRY / WIND VALLEY	Aeolian / past water	No permanent river	Qattara Depression, Egypt